Machine Learning Project Report

Earning 50K+? Income with a Fairness Check

Predicting Income Class (>50K / <=50K) on the Adult Census Dataset using Logistic Regression

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Project Title	Loan Approved or Not? (Income Classification)

1. Problem Statement

The goal is to predict whether an adult earns **more than $50K a year (>50K)** or **at most $50K (<=50K)** from census attributes such as age, education, marital status, occupation and weekly hours. This is a **binary classification** task. The notebook frames it as a "Loan Approved or Not?" screening problem: income level is a common proxy for repayment capacity, so a model that flags likely higher earners can support first-pass screening.

- Build a baseline Logistic Regression model and tune it (class weights, decision threshold).
- Handle the class imbalance (about 76% <=50K vs 24% >50K), where plain accuracy is misleading.
- Compare settings on Accuracy, Precision, Recall, F1, Specificity and AUC, then pick a final model by F1 (AUC as tie-breaker).

2. Dataset Description and Data Dictionary

The dataset is the **Adult (Census Income)** file, `adult.csv`: **32,561 rows x 15 columns** (6 numeric, 9 text/categorical). Missing values are stored as the text "?" (visible in `workclass` and `occupation`), so `isnull()` initially reports zero missing. The target column is `income`.

Column	Type	Description
age	Numeric	Age in years (17 to 90, mean 38.6)
workclass	Categorical	Employment sector (Private, Local-gov, Self-emp-inc, ...); "?" = missing
fnlwgt	Numeric	Census sampling weight (final weight)
education	Categorical	Highest education level (16 levels, e.g. HS-grad, Bachelors)
education.num	Numeric	Education level as a number (1 to 16)
marital.status	Categorical	Marital status (7 levels, e.g. Married-civ-spouse, Never-married)
occupation	Categorical	Job type (e.g. Prof-specialty, Exec-managerial); "?" = missing
relationship	Categorical	Role in household (6 levels, e.g. Husband, Own-child)
race	Categorical	Race (5 levels)
sex	Categorical	Male / Female
capital.gain	Numeric	Capital gains; zero for most people (median 0, mean 1,078)
capital.loss	Numeric	Capital losses; zero for most people (mean 87)
hours.per.week	Numeric	Hours worked per week (mean 40.4)
native.country	Categorical	Country of origin; "?" = missing
income	Target	Income class: <=50K (75.9%) or >50K (24.1%)
hours_over_40	Engineered	Hours above 40 per week: max(hours.per.week - 40, 0)

3. Data Cleaning and Exploratory Data Analysis

Cleaning and preparation steps

- Stripped extra spaces from text columns and converted "?" to missing values.
- Dropped **24 duplicate rows** (32,561 to **32,537** rows); no row had a missing target.
- Imputed missing values: **median** for numeric columns, **mode** for categorical columns. Zero missing values remained.
- Created `hours_over_40`, then one-hot encoded all text columns (`drop_first=True`), giving **98 input columns**.
- Split 80/20 with stratification (train 26,029 / test 6,508, `random_state=42`).
- Selected the **top 15 features** with SelectKBest (ANOVA F-score), then applied StandardScaler. Both were fitted on the training data only.

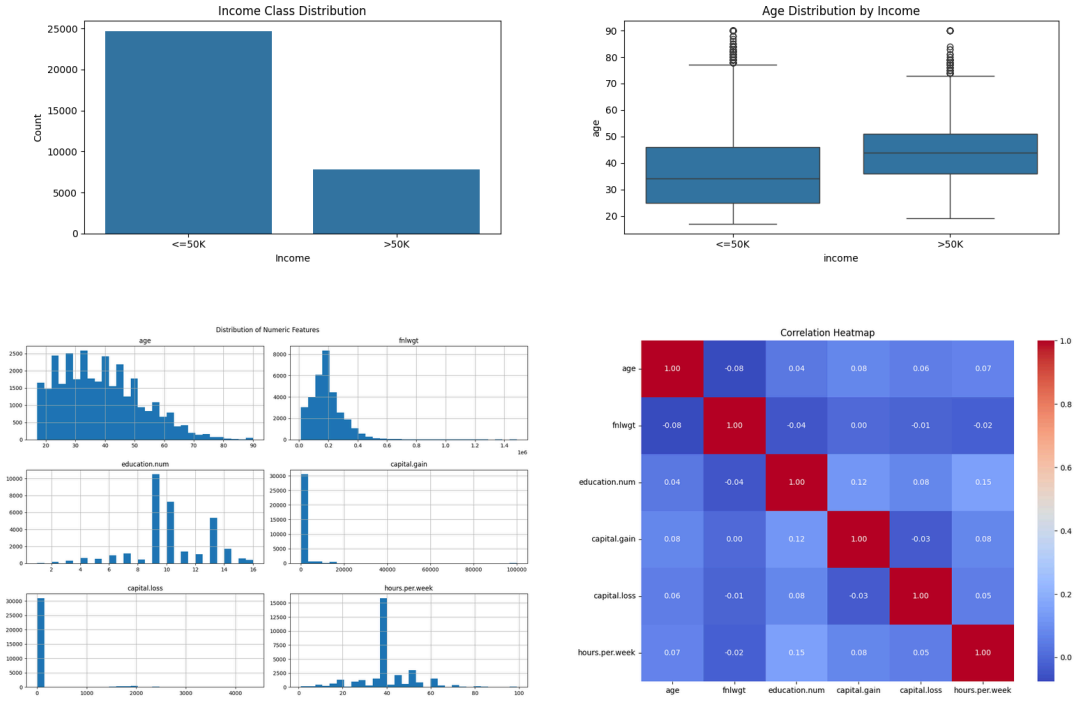


Figure 1. (a) Income class distribution, (b) age by income, (c) distribution of numeric features, (d) correlation heatmap of numeric features.

Key insights

- **Imbalanced target:** 75.9% earn <=50K and 24.1% earn >50K, so a model that always says <=50K would score about 76% accuracy. Recall, F1 and AUC matter more.
- **Age:** the >50K group is older (median about 44 vs about 34 for <=50K).
- **Skewed features:** `capital.gain` and `capital.loss` are zero for almost everyone; `hours.per.week` spikes at 40; `education.num` peaks at 9 (HS-grad), 10 and 13 (Bachelors).
- **Weak correlations:** the numeric features are only weakly correlated (largest is 0.15 between education.num and hours.per.week), so multicollinearity among them is not a concern. `fnlwgt` is nearly uncorrelated with everything.
- **Strongest predictors (F-score):** Married-civ-spouse (8,016), education.num (4,120), Never-married (3,667), age (1,885) and hours.per.week (1,812).

4. Model and Settings Tried

All models are **Logistic Regression** (`max_iter=1000`) trained on the 15 selected, standardised features and evaluated on the same 20% test set.

#	Setting	What changed
1	Baseline	Default Logistic Regression, class_weight=None, threshold 0.5
2	class_weight=None	Re-run after re-scaling; identical to the baseline (same model)
3	class_weight=balanced	Classes weighted inversely to frequency, so the minority class (>50K) counts more
4	Threshold 0.3 / 0.5 / 0.7	Probability cut-off varied on the unweighted model to see the precision-recall trade-off

Selection rule: highest F1-score, with AUC as tie-breaker. `class_weight=balanced` won (F1 0.662 vs 0.634) and was refitted as the **FINAL MODEL**.

5. Results

Model comparison (test set)

Setting	Accuracy	Precision	Recall	F1	Specificity	AUC
Baseline / class_weight=None	0.844	0.731	0.559	0.634	0.935	0.892
class_weight=balanced (FINAL)	0.791	0.542	0.849	0.662	0.773	0.892

Decision-threshold comparison (unweighted model)

Threshold	Precision	Recall
0.3	0.588	0.774
0.5	0.731	0.559
0.7	0.869	0.358

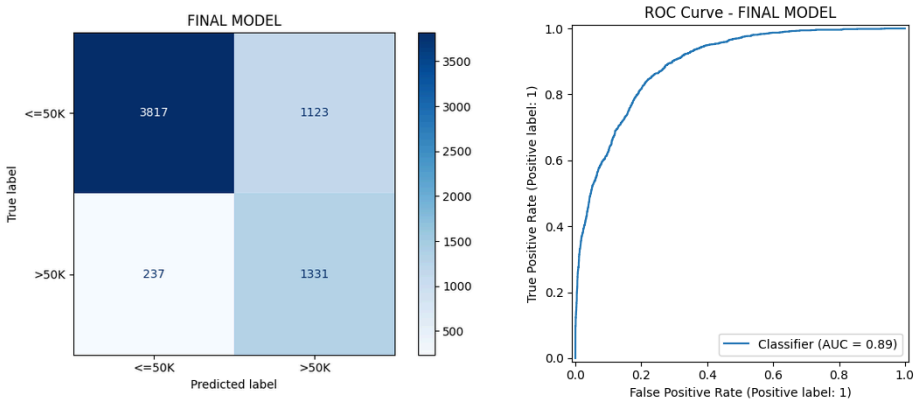


Figure 2. Final model: confusion matrix (left) and ROC curve, AUC = 0.892 (right).

Final confusion matrix: TN = 3,817, FP = 1,123, FN = 237, TP = 1,331. The model finds 1,331 of the 1,568 actual >50K cases (85%), but 1,123 <=50K people are wrongly flagged.

Coefficients (unweighted model): strongest positive are `capital.gain` (+2.20), Married-civ-spouse (+1.19) and `education.num` (+0.86); strongest negative are `relationship_Own-child` (-0.41), `hours_over_40` (-0.36) and `occupation_Other-service` (-0.30). On three sample rows the final model predicted <=50K with P(>50K) of 0.357, 0.194 and 0.204.

6. Conclusion and What I Learned

The final model is a **Logistic Regression with class_weight=balanced**: AUC 0.892, recall 0.849, F1 0.662, accuracy 0.791. It is the better choice when missing a genuine >50K case is costly. If false alarms are costlier, the unweighted model at a higher threshold is better (at 0.7, precision is 0.869 but recall falls to 0.358).

- **Accuracy can mislead.** The baseline had higher accuracy (0.844) yet missed 44% of high earners. F1, recall and the confusion matrix gave a truer picture.
- **AUC stayed at 0.892 in every setting.** Class weights and thresholds do not make the model rank people better; they only move the operating point along the precision-recall trade-off. The right choice depends on business cost.
- **Avoid data leakage.** Feature selection and scaling must be fitted on the training split only. In the notebook, SelectKBest was first run on the full data (exploration) and then re-run on the training data; the final pipeline used training data only, and the same 15 features were chosen.
- **Limitations:** one train/test split, no cross-validation, and only Logistic Regression was tested. `hours_over_40` has a negative coefficient, which is probably because it overlaps with `hours.per.week`. The data is census income, not real loan records, and it includes sensitive attributes (`sex_Male` is a selected feature), so a real lending use would need a fairness review.
- **Next steps:** try Decision Tree, Random Forest and Gradient Boosting, add cross-validation and hyper-parameter search, and audit performance across groups.